Sapienza University of Rome
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Francesco D'Amico

Research interests

- Learning rules, structure of data and training dynamics in deep learning
- Energy-based models and approximate likelihood methods
- Associative memories and attention-based architectures
- Theory of learning in generative models and neuroscience

Publications

2026 **Benchmarking Graph Neural Networks in Solving Hard Constraint Satisfaction Problems**
Skenderi G., Buffoni L., **D'Amico F.**, Machado D., Marino R., Negri M., Ricci-Tersenghi F., Lucibello C., Angelini M.C.
Accepted, Nature Communications

2026 **Implicit bias produces neural scaling laws in learning curves, from perceptrons to deep networks**
D'Amico F., Bocchi D. and Negri M.
The Fourteenth International Conference on Learning Representations - ICLR2026

2026 **Pseudo-likelihood produces associative memories able to generalize, even for asymmetric couplings**
D'Amico F., Bocchi D., Rossi S., Del Bono L. M. and Negri M.
Physica A: Statistical Mechanics and its Applications, doi:10.1016/j.physa.2026.131497

2025 **Statistical mechanics of vector Hopfield network near and above saturation**
Nicoletti F., **D'Amico F.** and Negri M.
J. Phys. A: Math. Theor. 58 505005, doi: 10.1088/1751-8121/ae2bd0

2025 **Algorithmic thresholds in combinatorial optimization depend on the time scaling**
Angelini M.C., Avila-González M., **D'Amico F.**, Machado D., Mulet R. and Ricci-Tersenghi F.
Phys. Rev. X, American Physical Society, doi:10.1103/dw9m-95vv

2024 **Self-attention as an attractor network: transient memories without back-propagation**
D'Amico F. and Negri M.
2024 IEEE Workshop on Complexity in Engineering (COMPENG), Florence, Italy, 2024, pp. 1-6, doi: 10.1109/COMPENG60905.2024.10741429.

Education

2023 - 2026 **PhD in Physics**, *Sapienza university of Rome.*
• Supervisors: Matteo Negri, Chiara Cammarota
• Theory of generative AI: attention-enabled models and associative memories (AMs) as a framework for generative abilities
• Most relevant courses: From Ergodicity breaking to algorithmic phase transitions (M. C. Angelini, G. Gradenigo), Statistical field theory (A. Cavagna), Foundations and Modern Applications of Random Matrix Theory (P. Vivo)

2021 – 2023 **Master's Degree in theoretical physics**, *Sapienza University of Rome.*
• Supervisors: Chiara Cammarota, Matteo Negri
• Thesis' title: "Random Features p-Spin Hopfield Model"
• Final degree mark: 110/110 cum laude
• Most relevant exams: Statistical mechanics and critical phenomena (E. Marinari), Statistical mechanics of disordered systems (F. Ricci Tersenghi), Statistical physics and machine learning (C. Cammarota), Advanced machine learning for physics (S. Giagu), Theoretical Biophysics (I. Giardina), Laboratory of Biophysics (E. Milanetti, G. Ruocco), Physics of complex systems (V. Loreto, F. Tria)

2021 – 2023 **Master's excellence program**, *Sapienza University of Rome.*
• Quantitative analysis of innovation levels in LLM's (CREF, SONY, V. Loreto)
• Numerical and experimental methods for the study of climate (A. M. Siani)
• Numerical methods for the study of complex biological systems (A. Giansanti)

2018 – 2021 **Bachelor's Degree in physics**, *Sapienza University of Rome.*
• Final mark: 110/110 cum laude
• Thesis' title: "Topological Vicsek model analyzed with machine learning techniques"
• Supervisor: S. Giagu

2018 – 2021 **Bachelor's excellence program**, *Sapienza University of Rome.*
• Computational Monte Carlo methods for the simulation of Hamiltonian systems (M. C. Angelini)
• Analytical and computational methods for the analysis of systems solvable by Green's method (D. Barducci)
• Theoretical study and computational simulation of active matter systems and analysis by machine learning methods (S. Giagu)

Other research experience

Visiting and periods abroad

2026 **Visiting PhD student at Johns Hopkins University**, *Latent dynamics in Large Language Models*, September 2026.
Supervisor: M. Wyart

2026 **Visiting PhD student at EPFL - PCSL laboratory**, *Hierarchical model of data for generative AI*, 01/03 - 29/05.
Supervisor: M. Wyart

2025 **Visiting PhD student at Complutense university of Madrid**, *Training of RBMs on power-law distributed random-features dataset*, 01/03 - 31/05.
Supervisor: B. Seoane

Talks

2026 **Understanding transformers requires new physical systems: pseudo-likelihood and vector-spin models**, *Seminar at Boston College*, 21/09/2026.

2026 **Implicit bias produces neural scaling laws in learning curves, from perceptrons to deep networks**, *Roccella conference on AI*, 01/09/2026.

2026 **Tutorial: how to build and train a GPT**, *Statistical Physics of Deep Learning III - Lake Como School of Advanced studies*, 16/06/2026.

2026 **Implicit bias produces scaling laws in learning curves, from perceptrons to deep networks** , *PCSL group meeting - École polytechnique fédérale de Lausanne (EPFL)*, 06/03/2026.

2026 **Scaling laws, from Perceptrons to Deep networks**, *4th workshop of Unione Matematica Italiana (UMI), Roma*, 23/01/2026.

2025 **Scaling laws, from Perceptrons to Deep networks**, *(1) Chimera journal club, Roma*, 07/10/2025, (2) Statistical mechanics group, University of Parma, 23/10/2025.

2025 **Statistical mechanics of autoregressive models: towards a theory of Self-Attention**, *PhD talks, Sapienza University, Roma*, 13/10/2025.

2024 **Self-attention as an attractor network: transient memories without back-propagation**, *SISSA journal club, Trieste*, 11/10/2024.

Conferences, workshops and schools

09/2026 **Simons Collaboration on the Physics of Learning and Neural Computation Annual Meeting**, *Simons foundation, New York, United States.*

09/2026 **Seminar at Boston College**, *Boston, Massachussets, United States*, seminar.

09/2026 **ROccella Conference on INference and AI**, *Roccella Ionica, Italy*, contributed talk .

06/2026 **Statistical Physics of Deep Learning III**, *Lake Como School of Advanced studies*, tutorial - notebook .

05/2026 **Fundamental principles of learning and representation: from Brains to LLMs**, *EPFL, Lausanne, Switzerland.*

04/2026 **The Fourteenth International Conference on Learning Representations - ICLR2026**, *Rio de Janeiro, Brazil*, poster .

04/2026 **New Frontiers in Associative Memories - NFAM2026 Workshop**, *Rio de Janeiro, Brazil*, poster .

01/2026 **4th Workshop of UMI Group, Italian Mathematics Union**, *Sapienza University, Roma, Italy*, contributed talk .

09/2025 **Summer school "Mathematical methods for high-dimensional data"**, *Sapienza University, Roma, Italy*, poster .

10/2025 **ROccella Conference on INference and AI**, *Roccella Ionica, Italy*, poster.

07/2025 **StatPhys29**, *Firenze, Italy*, poster.

07/2025 **Learning and optimization in high dimensions**, *Bocconi university, Milano, Italy*, poster.

06/2025 **Beg Rohu Summer school "Learning with machines, physics and minds"**, *Saint Pierre Quiberon, France*, poster.

10/2024 **ROccella Conference on INference and AI**, *Roccella Ionica, Italy*, poster.

07/2024 **Workshop on Complexity in Engineering COMPENG**, *Firenze, Italy*, poster.

07/2024 **Cargese Summer school "Complex and glassy systems"**, *Cargese, France*, poster.

Teaching and student supervision

2025 Co-supervisor of the Master's Degree thesis "Pseudo-likelihood associative memory on power-law distributed random-features dataset" of Sergio Manfrin, University of Milan

2025 Tutor for the course "Calculus 1" at the Department of Physics (Sapienza University) - 40 hours of exercises at the blackboard and Q&A sessions

2024 Tutor for the course "Calculus and statistics" at the Department of Biotechnology (Sapienza Universitu) - 40 hours of exercises at the blackboard and Q&A sessions

Languages

Mothertongue **Italian**

Advanced **English** *Level C1*

Intermediate **Spanish** *Level B1*

Basic **French** *Level A2*

Digital skills

Professional **Programming:** `Python, C, C++`.

Professional **Scientific packages:** `PyTorch, Numpy, Scipy, Matplotlib`.

Professional **Operating systems and HPC schedulers**: Linux, Windows, Slurm, PBS.

Intermediate **Scientific software:** `R, Wolfram Mathematica`

Other activities

2026 Selected as YAS Ambassador for FAO at World Food Forum 2026

2021-2026 Speaker and event curator for many editions of TEDxSapienzaU, the official TEDx organization in Sapienza university